VMware Cybersecurity Lab Setup

Week 1 Project Module 1

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Batch B084
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Project overview

This report documents the network configuration and connectivity validation of an existing Kali Linux virtual machine in VMware Workstation. The completed work establishes a NAT network, assigns a static IPv4 address to Kali, configures DNS and verifies internet access. Seven screenshots provide evidence of the configuration and test results.

The source lab brief uses VirtualBox. This implementation uses VMware Workstation with VMnet8 NAT. The scope of this report is the networking work demonstrated by the supplied evidence.

Network configuration

Component	Configured value
Virtual network	VMnet8 using NAT
Subnet	10.0.0.0/24
Subnet mask	255.255.255.0
Kali address	10.0.0.2/24
NAT gateway	10.0.0.254
DHCP range	10.0.0.128 to 10.0.0.200
DNS servers	1.1.1.1 and 8.8.8.8
Kali interface	eth0

The static Kali address lies outside the DHCP pool. The gateway uses a separate address, avoiding a conflict with Kali at 10.0.0.2.

1 VMware NAT network

VMnet8 is configured as NAT on subnet 10.0.0.0 with mask 255.255.255.0. The host virtual adapter and local DHCP service are enabled.

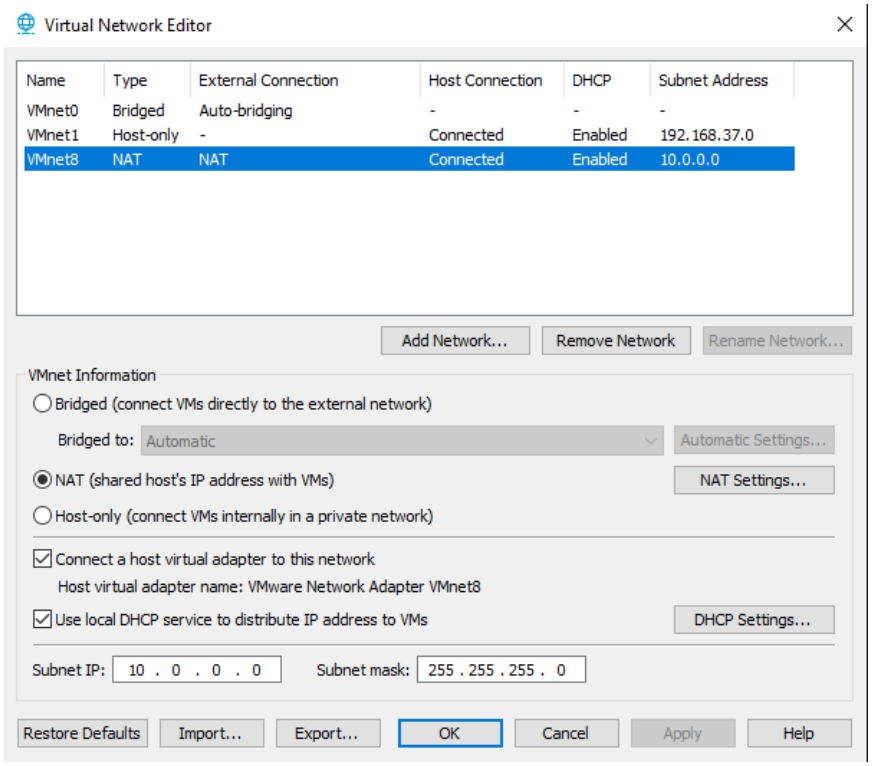


Figure 1. Virtual Network Editor showing VMnet8 NAT and the lab subnet.

The Kali virtual network adapter is connected to NAT, with Connected and Connect at power on enabled.

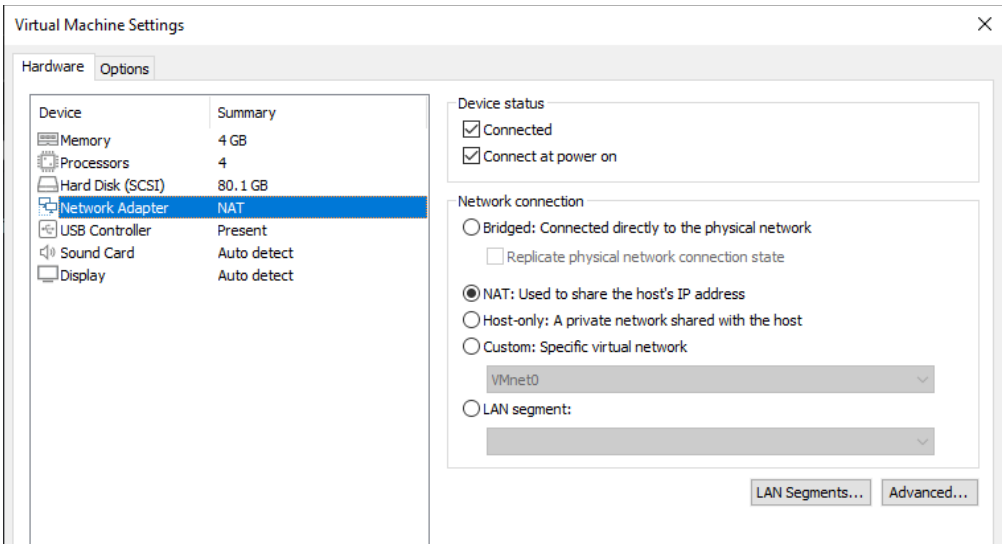


Figure 2. Kali virtual machine adapter using NAT.

2 Gateway and address allocation

The VMware NAT gateway is configured as 10.0.0.254. Kali uses this address as its default route for outbound traffic.

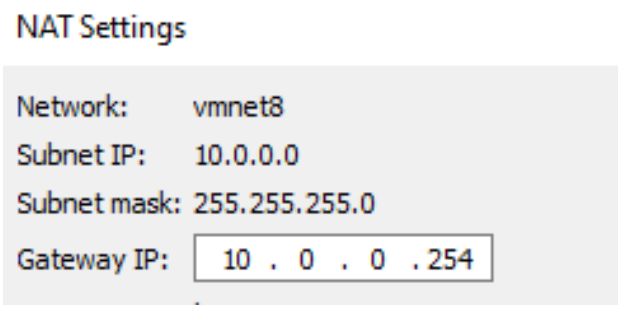


Figure 3. NAT Settings showing gateway 10.0.0.254.

The DHCP pool runs from 10.0.0.128 to 10.0.0.200. Kali uses the separate static address 10.0.0.2/24.

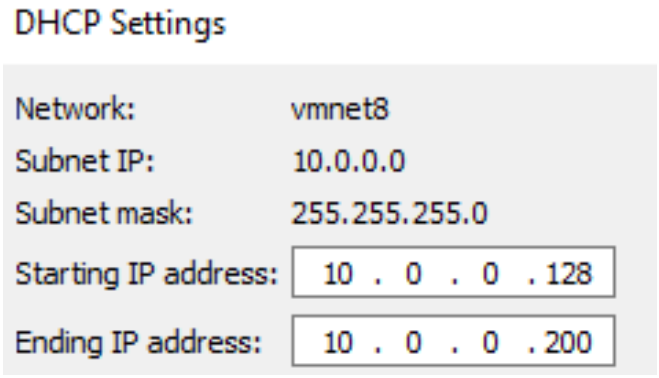


Figure 4. DHCP address pool configured on VMnet8.

Kali interface output confirms 10.0.0.2/24 on eth0. The routing table shows a default route via 10.0.0.254, and NetworkManager reports DNS servers 1.1.1.1 and 8.8.8.8.

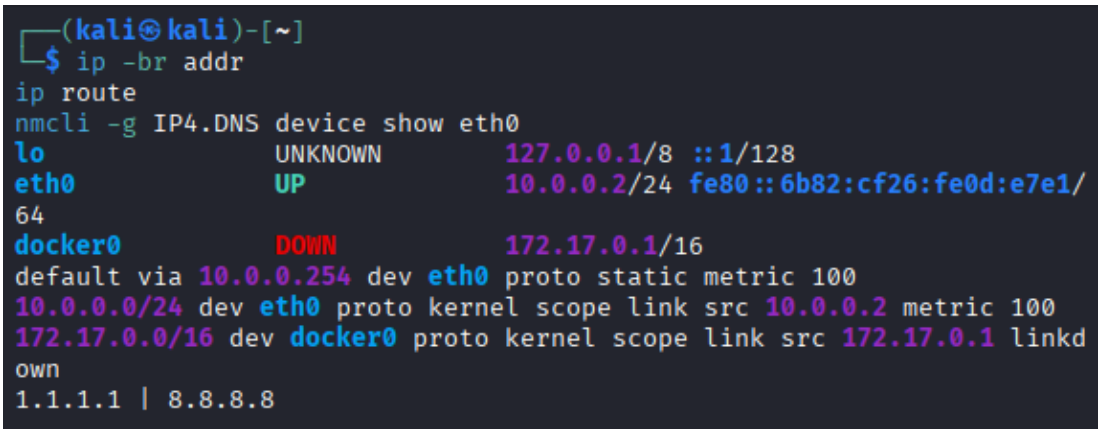


Figure 5. Kali interface address, default route and DNS configuration.

3 Connectivity test results

Connectivity was tested against the NAT gateway, an external IP address and a domain name. The gateway and 1.1.1.1 tests each returned four replies with zero packet loss. The google.com test resolved the domain and displayed four replies; its final summary is outside the screenshot.

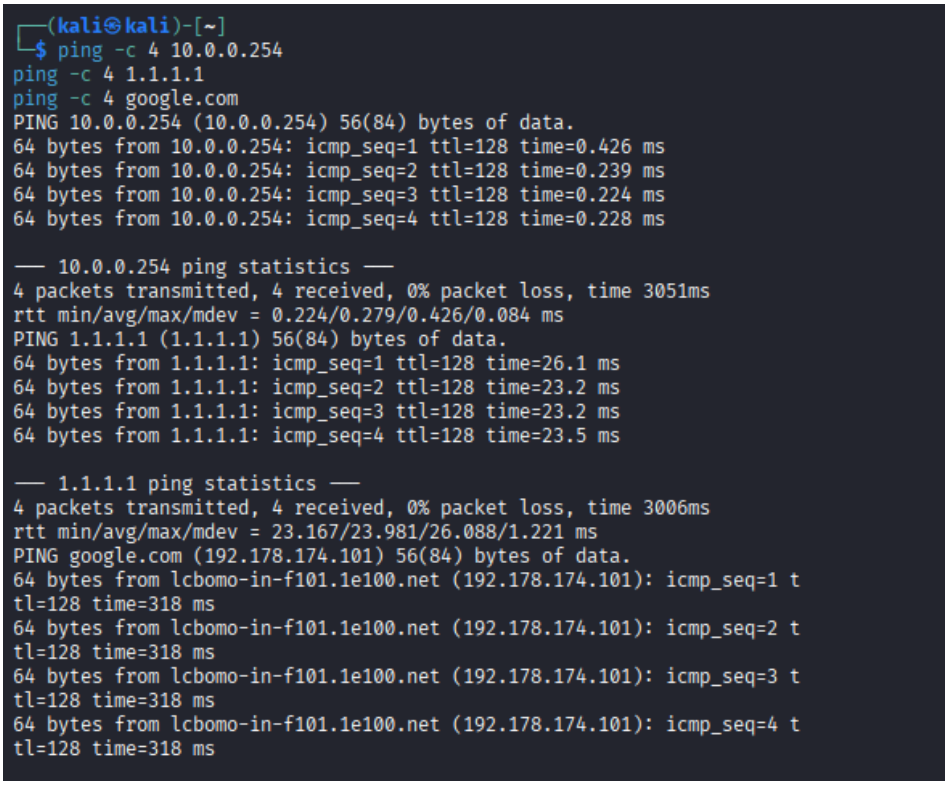


Figure 6. Gateway and external IP ping summaries and replies from google.com.

Test	Observed result
Gateway 10.0.0.254	4 replies; 0% packet loss
External address 1.1.1.1	4 replies; 0% packet loss
Domain google.com	DNS resolution and 4 replies visible

4 Web access and conclusion

Firefox in Kali successfully loaded https://www.kali.org. The visible HTTPS address and rendered website provide evidence of working web access from the virtual machine.

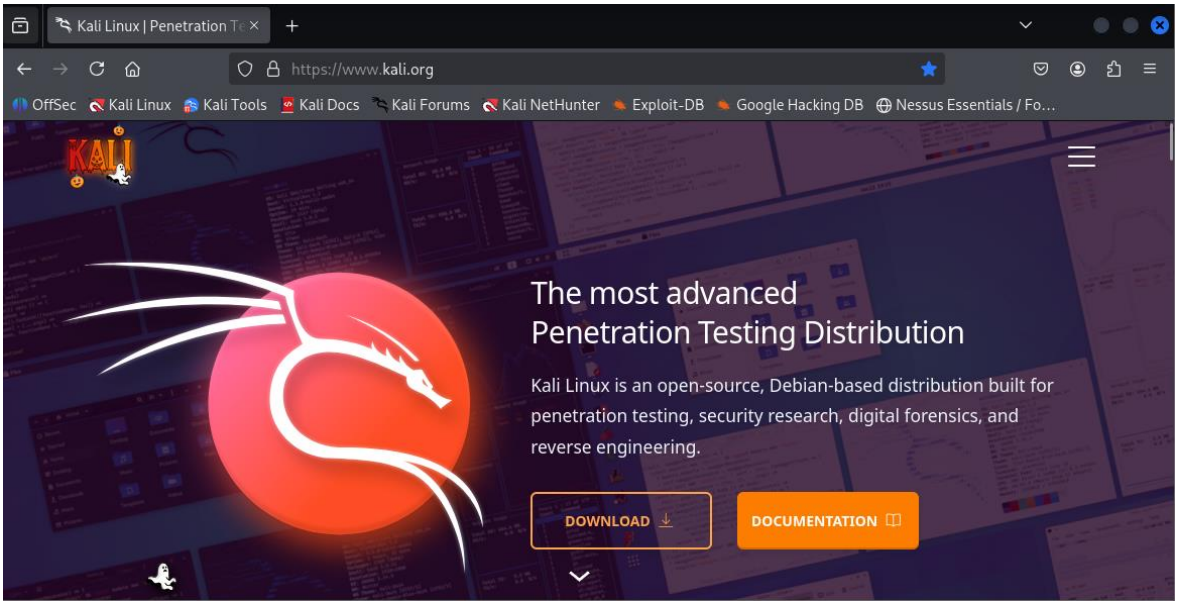


Figure 7. Kali website loaded in Firefox inside the virtual machine.

Conclusion

The documented networking work was completed successfully. VMware provides NAT connectivity on the required 10.0.0.0/24 subnet, and Kali uses the static address 10.0.0.2/24 with gateway 10.0.0.254. The supplied evidence confirms the configured DNS servers, gateway reachability, external connectivity, domain resolution and web access.

References

Networkwalks WK1 PM1 Lab Setup Virtualbox and Kali Linux v4.2 task brief.
Networkwalks Program Guide Workflows Cybersecurity Int v2.2b.
Configuration and test screenshots supplied by Israr M Qureshi.